

Nishant Kumar

PhD Scholar - Bioinformatics - Machine Learning

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Research Area

My research integrates **immunoinformatics**, **machine learning**, **LLMs**, and protein/peptide sequence analysis for rational subunit vaccine development. I have hands-on experience in large-scale dataset curation and **benchmarking** experimentally validated datasets to build predictive models for **B-cell epitopes**, **antibody-interacting residues**, **allergenicity**, **protein stability**, and **peptide half-life**. My work combines feature engineering and **LLM-based embeddings** with rigorous performance evaluation. I deploy these methods as freely available **web servers**, **pip packages**, **GitHub repositories**, and **standalone tools** for real-world use.

Education

Ph.D. in Computational Biology	July 2021 – Present
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India	
M.Sc. in Biotechnology	2019
Ch. Charan Singh University, Meerut, India	
B.Sc. in Biotechnology	2017
Ch. Charan Singh University, Meerut, India	

Research Experience

July 2021 -IIIT-Delhi, India

current *PhD Thesis:- Development of AI-based tools for designing subunit vaccines*

Advisor: Prof. Gajendra P.S. Raghava

Research: Curated and benchmarked experimental datasets; built deployable ML/DL pipelines; and applied state-of-the-art techniques (ensemble learning, protein language-model embeddings: ProtBERT/ESM) for tasks including B-cell epitope prediction, antifreeze proteins, protein thermostability, and Type 1 diabetes-related immunoinformatics.

Dec.2020 - Premas Biotech Pvt. Ltd, Manesar, Gurugram (Haryana)

June 2021 *Project:- “AXTEX-4D” in Mammalian Cell Culture Lab Department*

Supervisor: Dr. Ambica Baru

Responsibilities: Cell line maintenance, data analysis, 3D culture, Spheroids formation by using hanging drop method.

Jan.2019 - Institute Of Nanoscience & Technology, Mohali (Punjab)

July 2019 *Project:- Peptide-Based Theranostics In Treating Glioblastomas*

Supervisor: Dr. Jiban Jyoti Panda (Scientist ‘E’)

Responsibilities: Setting up instruments and laboratory equipment used to conduct and monitor experiments such as Centrifuges, Spectrophotometer, and Zeta sizer Nano ZSP.

Skills

Languages Python, R, Bash/Shell, SQL, PHP

Framework Pandas, Numpy, FastQC, Scikit-Bio, Scikit-learn, tensorflow, Matplotlib, Seaborn, Pytorch

Bioinformatics Docker, CD-Hit, BLAST, MEME/MAST, Merci

Language Models ProtBert, PeptideBert, ESM Fold, ESM2, ProtXL-T5

Platforms Pycharm, Jupyter, Visual studio code, Spyder, Linux, XAMPP

Wet Lab Cell/Tissue culture, Spheroid Formation, Data collection, Drug Treatment

Instruments UV-Vis spectrophotometer, Zetasizer Nano ZSP, Bright Field Microscope, FACS (Fluorescence-activated cell sorting), Centrifuge Machine, Laminar Air Flow

Publications

Peer-reviewed journal

- **HAIRPred: Prediction of antibody-interacting residues**
R. Sahni, **Nishant Kumar**, Raghava GPS Protein Science, 2025
- **Prediction of anti-freezing proteins from their evolutionary profile**
Kumar N, Choudhury S, Bajiya N, Patiyal S, Raghava GPS Proteomics, 2024
- **CLBTope: Predicting of linear and conformational B-cell epitopes from sequence**
Kumar N, Tripathi S, Sharma N, Patiyal S, Devi NL, Raghava GPS Computers in Biology and Medicine, 2024
- **Multi-perspectives and challenges in identifying B-cell epitopes**
Kumar N, Bajiya N, Patiyal S, Raghava GPS Protein Science, 2023
- **DMPPred: identification of antigenic regions responsible for inducing T1 diabetes mellitus**
Kumar N, Patiyal S, Choudhury S, Tomer R, Dhall A, Raghava GPS Briefings in Bioinformatics, 2023
- **Prediction and design of thermostable proteins with a desired melting temperature**
Tijare P, **Kumar N**, Raghava GPS (*Joint First Author) Scientific Reports, 2025
- **Prediction of inhibitory peptides against *E. coli* with desired MIC value**
Bajiya N, **Kumar N**, Raghava GPS Scientific Reports, 2025
- **Prediction of hemolytic peptides and their hemolytic concentration**
Rathore AS, **Kumar N**, Choudhury S, Mehta NK, Raghava GPS Communications Biology, 2025
- **PlantDRPpred: Prediction of plant resistance proteins**
Gahlot PS, Choudhury S, Bajiya N, **Kumar N**, Raghava GPS Proteomics, 2025
- **HLA-DR4Pred2: An improved method for predicting HLA-DRB1*04:01 binders**
Patiyal S, Dhall A, **Kumar N**, Raghava GPS Methods, 2024
- **Dual blood-brain barrier-glioma targeting peptide-poly(levodopamine) hybrid nanoplat-forms as potential near infrared phototheranostic agents in glioblastoma**
Dube T, **Kumar N**, Bishnoi M, Panda JJ Bioconjugate Chemistry, 2021

Preprints

- **MAP format for representing chemical modifications, annotations, and mutations in protein sequences: An extension of the FASTA format**
Shendre A, Mehta NK, Rathore AS, **Kumar N**, Patiyal S, Raghava GPS arXiv (q-bio.BM), 2025
- **In silico tool for identification of colorectal cancer from cell-free DNA biomarkers**
Mathur K, Jain S, Bajiya N, **Kumar N**, Raghava GPS arXiv (q-bio.GN), 2025
- **PEPlife2: An updated repository of the half-life of peptides**
Alam U, Chaudhary K, **Kumar N**, Tomer R, Patiyal S, Raghava GPS bioRxiv, 2025
- **CBTOPE2: An improved method for predicting of conformational B-cell epitopes**
Pandey A, Megha, **Kumar N**, Sahni R, Raghava GPS arXiv (q-bio.BM), 2025

Teaching experience

Teaching Assistant, IIIT-Delhi, India

- Machine Learning in Biomedical Applications 2022
- Big Data Mining in Healthcare 2023

Key Responsibilities

- Prepared and deliver tutorials, graded assignments and provide constructive feedback to students
- Mentored bachelor's and master students on research projects

Conference and Achievements

1. **Presented - DMPPRed at JBI2025 XV Bioinformatics Symposium (JBI2025)** at Madrid, Spain, 22 – 24 October 2025
2. **Presented - Multi-perspectives and challenges in identifying B-cell epitopes** at IIIT-Delhi (RIISE), 15-16 March, 2024
3. **Adipo-Cible Winter School — Obesity Unraveled: From Cellular Mechanisms to Therapeutic Pathways** held by Université Côte d’Azur, France, Dec. 16–19, 2024.
4. **InBix–ADNAT 2024 Conference:** Integrating Archaeogenetics, Forensics & Multi-omics for Human Health Breakthroughs, 28–30 November 2024, Banaras Hindu University, Varanasi, India.
5. **3-day Hands-on Workshop:** “Introductory Omics and RNA-seq Data Analysis,” 11–13 July 2024, organised by Bio-IT Centre, IBAB, Bengaluru, India.
6. **Workshop:** Establishing and Sustaining Cloud Infrastructure for Informatics-based Healthcare, 22–23 March 2024, IIIT-Delhi, India.
7. **CSIR-UGC JRF Fellowship (2020)**

Referees

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